

《数据库系统原理》实验报告（4）					
题目：SQL 综合实验					
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实验环境： 1. 操作系统：Windows 11 2. 容器平台：Docker Desktop for Windows（版本：4.65.0） 3. 数据库镜像：oceanbase/oceanbase-ce:latest 4. 数据库系统：OceanBase Community Edition 4.3.5.5					
1、建表（见附录一），表内字段的类型可以自行定义（合理即可），注意建表时不要忽略各表的主键约束和表间的外键约束； <pre> obclient(root@sys)[(none)]> create database BookManagement; Query OK, 1 row affected (2.527 sec) obclient(root@sys)[(none)]> use BookManagement; Database changed obclient(root@sys)[BookManagement]> create table Book (-> Book_no varchar(10) primary key, -> Book_name varchar(30), -> Author varchar(30), -> Price float ->); Query OK, 0 rows affected (1.041 sec) obclient(root@sys)[BookManagement]> create table Student (-> Student_no varchar(10) primary key, -> Student_name varchar(30), -> Grade varchar(10) ->); Query OK, 0 rows affected (0.302 sec) obclient(root@sys)[BookManagement]> create table Borrow (-> Student_no varchar(10), -> Book_no varchar(10), -> Return_date datetime, -> primary key(Student_no, Book_no), -> foreign key(Student_no) references Student(Student_no) on delete cascade, -> foreign key(Book_no) references Book(Book_no) on delete cascade ->); Query OK, 0 rows affected (0.851 sec) </pre>					
2、插入样例数据（见附录二）；					

```
obclient(root@sys)[BookManagement]> insert into Book values
-> ('T1001','Java 程序设计','李新珊',89.5),
-> ('T1002','数据库原理及应用','王敏',39),
-> ('T1003','Java 高级编程','陈明海',63.5),
-> ('T1004','专业英语','张悦宁',23.1),
-> ('T1005','C++程序设计','马天颖',83.2),
-> ('T1006','编译原理','王鑫单',65);
Query OK, 6 rows affected (0.098 sec)
Records: 6 Duplicates: 0 Warnings: 0

obclient(root@sys)[BookManagement]> insert into Student values
-> ('K005','张鑫翁','大一'),
-> ('K003','徐晨皓','大二'),
-> ('K002','王三优','大三'),
-> ('K001','刘孔明','大三'),
-> ('K004','吴宇涵','大四');
Query OK, 5 rows affected (0.014 sec)
Records: 5 Duplicates: 0 Warnings: 0

obclient(root@sys)[BookManagement]> insert into Borrow values
-> ('K001','T1006','2023-10-9'),
-> ('K001','T1001','2024-3-1'),
-> ('K002','T1002','2023-10-9'),
-> ('K002','T1003','2024-4-5'),
-> ('K002','T1001','2023-11-3'),
-> ('K003','T1005','2024-1-4'),
-> ('K004','T1002','2024-2-5');
Query OK, 7 rows affected (0.041 sec)
```

3、查询书名中包含“程序设计”的图书信息，输出所有信息（包括书名、书号、作者、单价），并按照单价降序排列；

```
obclient(root@sys)[BookManagement]> select * from Book where Book_name like "%程序设计%" order by Price desc;
+-----+-----+-----+-----+
| Book_no | Book_name      | Author | Price |
+-----+-----+-----+-----+
| T1001   | Java 程序设计 | 李新珊 | 89.5  |
| T1005   | C++程序设计   | 马天颖 | 83.2  |
+-----+-----+-----+-----+
2 rows in set (0.393 sec)
```

4、查询借阅了书名为“数据库原理及应用”的学生信息，输出该学生的学号、姓名和年级，并按照学号升序排列；

```
obclient(root@sys)[BookManagement]> select s.Student_no, s.Student_name, s.Grade from Borrow, Student as s, Book where Borrow.Student_no = s.Student_no and Borrow.Book_no = Book.Book_no and Book.Book_name = "数据库原理及应用" order by s.Student_no asc;
+-----+-----+-----+
| Student_no | Student_name | Grade |
+-----+-----+-----+
| K002       | 王三优       | 大三  |
| K004       | 吴宇涵       | 大四  |
+-----+-----+-----+
```

5、统计每个学生借书信息，输出每个学生的学号、借书书名和还书日期；

```
obclient(root@sys)[BookManagement]> select Student.Student_no, Book.Book_name, Borrow.Return_date
-> from Book natural join Student natural join Borrow;
+-----+-----+-----+
| Student_no | Book_name          | Return_date      |
+-----+-----+-----+
| K001       | Java 程序设计      | 2024-03-01 00:00:00 |
| K002       | Java 程序设计      | 2023-11-03 00:00:00 |
| K002       | 数据库原理及应用  | 2023-10-09 00:00:00 |
| K004       | 数据库原理及应用  | 2024-02-05 00:00:00 |
| K002       | Java 高级编程      | 2024-04-05 00:00:00 |
| K003       | C++程序设计        | 2024-01-04 00:00:00 |
| K001       | 编译原理           | 2023-10-09 00:00:00 |
+-----+-----+-----+
7 rows in set (0.044 sec)
```

6、查询所有借阅已过期图书的信息，输出学生学号、姓名、书名和还书日期，并按还书日期降序排列；

```
obclient(root@sys)[BookManagement]> select Student.Student_no, Student.Student_name, Book.Book_name, Borrow.Return_date
-> from Borrow natural join Student natural join Book
-> where Return_date < '2026-04-20'
-> order by Return_date desc;
+-----+-----+-----+-----+
| Student_no | Student_name | Book_name          | Return_date      |
+-----+-----+-----+-----+
| K002       | 王三优       | Java 高级编程      | 2024-04-05 00:00:00 |
| K001       | 刘孔明       | Java 程序设计      | 2024-03-01 00:00:00 |
| K004       | 吴宇涵       | 数据库原理及应用  | 2024-02-05 00:00:00 |
| K003       | 徐晨皓       | C++程序设计        | 2024-01-04 00:00:00 |
| K002       | 王三优       | Java 程序设计      | 2023-11-03 00:00:00 |
| K002       | 王三优       | 数据库原理及应用  | 2023-10-09 00:00:00 |
| K001       | 刘孔明       | 编译原理           | 2023-10-09 00:00:00 |
+-----+-----+-----+-----+
7 rows in set (0.040 sec)
```

7、查询没有借阅过书的学生信息，输出学生姓名和学号；

```
obclient(root@sys)[BookManagement]> select s.Student_name, s.Student_no
-> from Student as s
-> where s.Student_no not in(
-> select s1.Student_no
-> from (Student as s1) natural join Borrow);
+-----+-----+
| Student_name | Student_no |
+-----+-----+
| 张鑫翁       | K005       |
+-----+-----+
1 row in set (0.017 sec)
```

8、查询借了"Java 程序设计"但没有借"数据库原理及应用"的读者信息，输出这些学生的学号，并按照学号升序排列；

```
obclient(root@sys)[BookManagement]> create view FullInfo as(
-> select * from Book natural join Student natural join Borrow);
Query OK, 0 rows affected (0.407 sec)
```

```
obclient(root@sys)[BookManagement]> desc FullInfo;
```

Field	Type	Null	Key	Default	Extra
Book_no	varchar(10)	NO		NULL	
Student_no	varchar(10)	NO		NULL	
Book_name	varchar(30)	YES		NULL	
Author	varchar(30)	YES		NULL	
Price	float	YES		NULL	
Student_name	varchar(30)	YES		NULL	
Grade	varchar(10)	YES		NULL	
Return_date	datetime	YES		NULL	

8 rows in set (0.014 sec)

```
obclient(root@sys)[BookManagement]>
Student_no from FullInfo where Book_name = "数据库原理及应用")) order by s.Student_no asc;no in ((select Student_no from FullInfo where Book_name = "Java 程序设计") except (select
Student_no from FullInfo where Book_name = "数据库原理及应用")) order by s.Student_no asc;
1 row in set (0.003 sec)
```

被截断的完整语句为: select s.Student_no from Student as s where s.Student_no in ((select Student_no from FullInfo where Book_name = "Java 程序设计") except (select Student_no from FullInfo where Book_name = "数据库原理及应用")) order by s.Student_no asc;

9、创建一个过程，使之能够实现如下功能：

10、修改借阅表，增加字段“借阅状态”（字段名为“Borrow_state”），字段含义为表示图书的借阅状态是否已经过期；

11、并根据表中已有数据为该字段赋值（所赋的值与表定义时的数据类型保持一致即可，比如可以定义已到期图书的“借阅状态”为 True，未到期图书的“借阅状态”为 False），要求使用 if 语句进行条件判断；

```

obclient(root@sys)[BookManagement]> delimiter $$
obclient(root@sys)[BookManagement]> create procedure update_bstate()
-> begin
->     declare done int default 0;
->     declare v_student_no varchar(10);
->     declare v_book_no varchar(10);
->     declare v_return_date datetime;
->
->     declare cur cursor for
->         select Student_no, Book_no, Return_date
->         from Borrow;
->
->     declare continue handler for not found set done = 1;
->
->     open cur;
->
->     read_loop: loop
->         fetch cur into v_student_no, v_book_no, v_return_date;
->
->         if done = 1 then
->             leave read_loop;
->         end if;
->
->         if v_return_date < '2026-04-20' then
->             update Borrow
->             set Borrow_state = true
->             where Student_no = v_student_no
->             and Book_no = v_book_no;

```

```

->         else
->             update Borrow
->             set Borrow_state = false
->             where Student_no = v_student_no
->             and Book_no = v_book_no;
->         end if;
->     end loop;
->
->     close cur;
-> end$$

```

Query OK, 0 rows affected (0.483 sec)

```

obclient(root@sys)[BookManagement]> delimiter ;

```

```
obclient(root@sys)[BookManagement]> select * from Borrow;
+-----+-----+-----+-----+
| Student_no | Book_no | Return_date          | Borrow_state |
+-----+-----+-----+-----+
| K001       | T1001   | 2024-03-01 00:00:00 | NULL        |
| K001       | T1006   | 2023-10-09 00:00:00 | NULL        |
| K002       | T1001   | 2023-11-03 00:00:00 | NULL        |
| K002       | T1002   | 2023-10-09 00:00:00 | NULL        |
| K002       | T1003   | 2024-04-05 00:00:00 | NULL        |
| K003       | T1005   | 2024-01-04 00:00:00 | NULL        |
| K004       | T1002   | 2024-02-05 00:00:00 | NULL        |
+-----+-----+-----+-----+
7 rows in set (0.014 sec)

obclient(root@sys)[BookManagement]> call update_bstate();
Query OK, 1 row affected (2.007 sec)

obclient(root@sys)[BookManagement]> select * from Borrow;
+-----+-----+-----+-----+
| Student_no | Book_no | Return_date          | Borrow_state |
+-----+-----+-----+-----+
| K001       | T1001   | 2024-03-01 00:00:00 | 1           |
| K001       | T1006   | 2023-10-09 00:00:00 | 1           |
| K002       | T1001   | 2023-11-03 00:00:00 | 1           |
| K002       | T1002   | 2023-10-09 00:00:00 | 1           |
| K002       | T1003   | 2024-04-05 00:00:00 | 1           |
| K003       | T1005   | 2024-01-04 00:00:00 | 1           |
| K004       | T1002   | 2024-02-05 00:00:00 | 1           |
+-----+-----+-----+-----+
7 rows in set (0.002 sec)
```

此处约定未到期的书 Borrow_state 为 True。

出现的问题：

9-11 题，OceanBase 不支持课件里的 for r as select .. from 语法

解决方案:

改用游标 `cursor` 语法